



(+91)-9046408944
abhishekroy@rnd.iitg.ac.in
Coochbehar, West Bengal, India
Department of EEE, Indian Institute of Technology Guwahati
<https://orcid.org/0000-0002-5301-9157>
<https://www.researchgate.net/profile/Abhishek-Roy-6>
<https://www.linkedin.com/in/dr-abhishek-roy-74983018a/>



ABHISHEK ROY

Research Overview

My current research interests are sequence design and error-correcting codes for applications in wireless communication and information theory. For construction, I use tools such as Boolean functions, combinatorics, and discrete mathematics. Also, some basic knowledge of programming languages, like C/C++, or software tools, such as MATLAB, can be helpful. The open problems in this field require a broad range of mathematical tools to solve. My objective is to explore such problems related to communication and information theory using novel mathematical frameworks.

Employment

Oct, 2024- Present	National Post-Doctoral Fellow (NPDF) Department of EEE · Indian Institute of Technology, Guwahati
Nov, 2024- Sept, 2026	Institute Post-Doctoral Fellow (IPDF) Department of EEE · Indian Institute of Technology, Guwahati

Ph.D. Thesis

Title	CONSTRUCTION OF 1-D SEQUENCES AND 2-D ARRAYS WITH FLEXIBLE LENGTHS FOR APPLICATION IN WIRELESS COMMUNICATION
Defense	12 April, 2024
Supervisor	Dr. Sudhan Majhi, Department of Electrical Communication Engineering, IISc, Bangalore, email: smajhi@iisc.ac.in
Co-supervisor	Dr. Subhabrata Paul, Department of Mathematics, IIT Patna, email: subhabrata@iitp.ac.in

Education

- 2024 **Doctor of Philosophy**
Mathematics · Indian Institute of Technology, Patna
- 2016 **Master of Science**
Mathematics · Indian Institute of Technology, Kanpur
- 2014 **Bachelor of Science**
Mathematics · Jadavpur University, Kolkata
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Programming Skills

1. $\text{\LaTeX}$
 2. MATLAB
 3. C/C++
 4. Mathematica
 5. Python
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Fellowships and Awards

- 2020–2023 CSIR-Senior Research Fellowship
- 2018–2020 CSIR-Junior Research Fellowship
- 2011–2016 INSPIRE-DST Scholarship for Top 1% in XII Standard
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Competitive Examinations

1. Joint CSIR-UGC NET in *Mathematical Science* (2016)
 2. GATE in *Mathematics* (2021)
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Journal Publications

- 2025 | P. Priyanshu, A. Roy, S. Paul, and S. Majhi, "Constructions of non-binary Golay complementary pairs of new lengths", *Cryptography and Communications*, vol. 17, pp. 809-822, doi: <https://doi.org/10.1007/s12095-025-00782-x>
- 2024 | A. Roy and S. Majhi, "Lower bounds on the maximum cross-correlations of 2-D quasi-complementary array sets", *Cryptography and Communications*, vol. 16, no. 2, pp. 229-247, doi: <https://doi.org/10.1007/s12095-023-00665-z>
- 2021 | A. Roy, P. Sarkar, and S. Majhi, "A direct construction of q-ary 2-D Z-complementary array pair based on generalized Boolean functions", *IEEE Communications Letters*, vol. 25, no. 3, pp. 706-710, doi: <https://doi.org/10.1109/LCOMM.2020.3038726>
- 2020 | P. Sarkar, A. Roy, and S. Majhi, "Construction of Z-complementary code sets with non-power-of-two lengths based on generalized Boolean functions", *IEEE Communications Letters*, vol. 24, no. 8, pp. 1607-1611, doi: <https://doi.org/10.1109/LCOMM.2020.2987625>
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Conference Publications

- 2025 | A. Roy, S. Das, and S. Majhi, "Construction of MOGCS with Flexible Parameters via Extended Boolean Functions", In *IEEE International Symposium on Information Theory (ISIT)*, Ann Arbor, Michigan, USA, pp. 1-6, doi: <https://doi.org/10.1109/ISIT63088.2025.11195398>
- 2024 | A. Roy, S. Majhi, and S. Paul, "Systematic Construction of Golay Complementary Sets of Arbitrary Lengths and Alphabet Sizes", In *IEEE Information Theory Workshop (ITW)*, Shenzhen, China, pp. 430-435, doi: <https://doi.org/10.1109/ITW61385.2024.10806962>
- 2023 | A. Roy and S. Majhi, "Direct constructions of multiple shift complementary sets of flexible lengths", In *IEEE International Symposium on Information Theory (ISIT)*, pp. 2440-2445, doi: <https://doi.org/10.1109/ISIT54713.2023.10206672>
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Articles in Preparation

- 2025 | A. Roy, "Construction of 2-D Z-Complementary Array Code Sets with Flexible Lengths for Different System Requirements", [Online] <https://arxiv.org/abs/2109.00970>
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Scientific Affiliations

- AMS (Introductory Member)
 - IEEE (Member)
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Languages

1. Bengali (*Mother Tongue*)
 2. English (*Professional*)
 3. Hindi (*Intermediate*)
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Extra Curricular

- Playing Guitar
 - Drawing
 - Quizzing
 - Photography
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